

Case Study: Engel Curves and Quantile Regression

Workshop Exercise

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1 Learning Objectives

By the end of this exercise you will be able to:

- Fit quantile regression models using the Engel dataset
- Compare OLS and quantile regression estimates
- Examine how the relationship between income and food expenditure varies across quantiles
- Construct confidence intervals using `se = "nid"`
- Interpret a coefficient process plot
- Write up quantile regression findings in plain language

2 Introduction

A classic question in economics is:

How does household food expenditure change as income increases?

The Engel dataset is one of the most famous examples used in the quantile regression literature and is included in the `quantreg` package.

Unlike the CPS1988 wage example, this dataset contains only two variables:

- Household income
- Household food expenditure

This allows us to focus entirely on understanding quantile regression.

2.1 Exercise 1: Exploring the Data

Inspect the dataset

```
glimpse(engel)
```

Rows: 235

Columns: 2

```
$ income <dbl> 420.1577, 541.4117, 901.1575, 639.0802, 750.8756, 945.7989, 82~
```

```
$ foodexp <dbl> 255.8394, 310.9587, 485.6800, 402.9974, 495.5608, 633.7978, 63~
```

```
summary(engel)
```

income	foodexp
Min. : 377.1	Min. : 242.3
1st Qu.: 638.9	1st Qu.: 429.7
Median : 884.0	Median : 582.5
Mean : 982.5	Mean : 624.2
3rd Qu.: 1164.0	3rd Qu.: 743.9
Max. : 4957.8	Max. : 2032.7

Questions

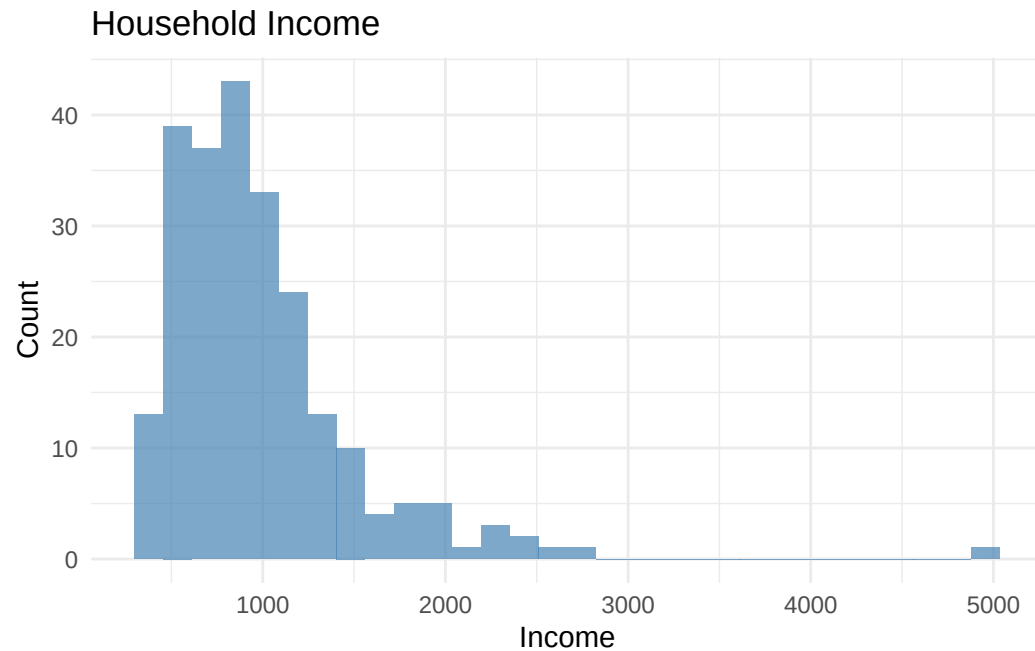
1. How many households are included?
2. Which variable is the outcome?
3. Which variable is the predictor?

Answers

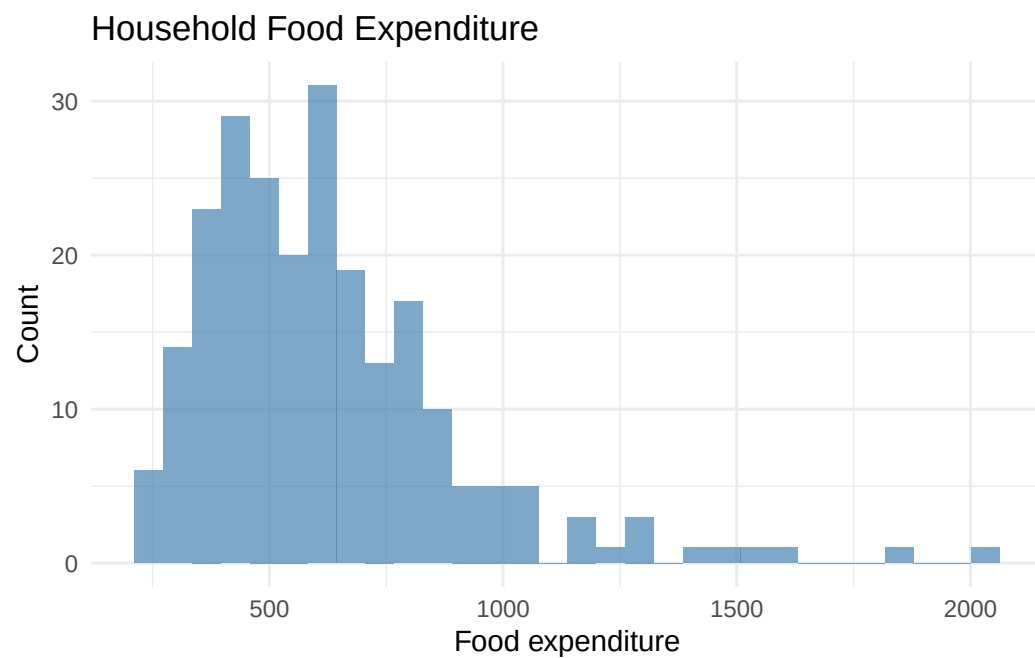
1. The dataset contains 235 households.
2. The outcome variable is `foodexp` (household food expenditure).
3. The predictor variable is `income` (household income).

2.2 Visualising the Variables

```
ggplot(engel, aes(x = income)) +  
  geom_histogram(bins = 30,  
                 fill = "steelblue",  
                 alpha = 0.7) +  
  labs(  
    title = "Household Income",  
    x = "Income",  
    y = "Count"  
  )
```



```
ggplot(engel, aes(x = foodexp)) +  
  geom_histogram(bins = 30,  
                 fill = "steelblue",  
                 alpha = 0.7) +  
  labs(  
    title = "Household Food Expenditure",  
    x = "Food expenditure",  
    y = "Count"  
  )
```



Questions

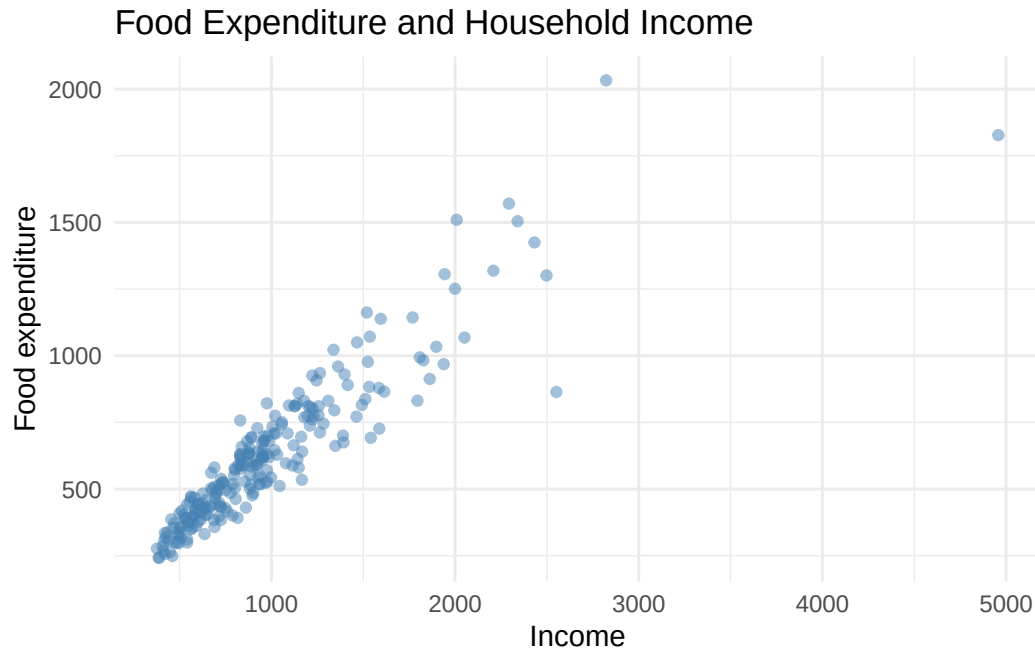
1. Is income skewed?
2. Is food expenditure skewed?
3. Why might averages be insufficient to describe these distributions?

Answers

1. Income is positively skewed, with a small number of relatively high-income households.
2. Food expenditure is also positively skewed, although less strongly than income.
3. Averages may hide important differences across the distribution. Households with the same average expenditure may have very different spending patterns at the lower and upper tails.

2.3 Exercise 2: Income and Food Expenditure

```
ggplot(
  engel,
  aes(
    x = income,
    y = foodexp
  )
) +
  geom_point(alpha = 0.5,
             colour = "steelblue") +
  labs(
    title = "Food Expenditure and Household Income",
    x = "Income",
    y = "Food expenditure"
  )
)
```



Questions

1. Is the relationship positive?
2. Does variability appear constant across the income range?
3. Why might quantile regression be useful here?

Answers

1. Yes. Higher-income households generally spend more on food.
2. No. The spread of food expenditure increases with income, suggesting heteroscedasticity.
3. Quantile regression allows us to examine whether the effect of income differs for households with low versus high food expenditure.

2.4 Exercise 3: OLS Baseline

```
ols_fit <- lm(
  foodexp ~ income,
  data = engel
)

summary(ols_fit)
```

Call:

```
lm(formula = foodexp ~ income, data = engel)
```

Residuals:

Min	1Q	Median	3Q	Max
-725.70	-60.24	-4.32	53.41	515.77

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	147.47539	15.95708	9.242	<2e-16 ***
income	0.48518	0.01437	33.772	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 114.1 on 233 degrees of freedom

Multiple R-squared: 0.8304, Adjusted R-squared: 0.8296

F-statistic: 1141 on 1 and 233 DF, p-value: < 2.2e-16

Questions

1. What is the coefficient on income?
2. Interpret the coefficient.
3. What does OLS tell us about the relationship between income and food expenditure?

Answers

1. The coefficient on income is approximately 0.485.
2. A one-unit increase in household income is associated with approximately 0.485 additional units of food expenditure, on average.
3. OLS indicates a strong positive relationship between income and food expenditure: households with higher incomes tend to spend more on food. However, OLS summarises this relationship using a single average effect and does not tell us whether the income effect differs across the food expenditure distribution.

2.5 Exercise 4: Median Regression

```
fit_median <- rq(
  foodexp ~ income,
  tau = 0.50,
```

```

data = engel
)

summary(
  fit_median,
  se = "nid"
)

```

Call: rq(formula = foodexp ~ income, tau = 0.5, data = engel)

tau: [1] 0.5

Coefficients:

	Value	Std. Error	t value	Pr(> t)
(Intercept)	81.48225	19.25066	4.23270	0.00003
income	0.56018	0.02828	19.81032	0.00000

```

# Alternative:
# summary(
#   fit_median,
#   se = "boot",
#   R = 1000
# )

```

Questions

1. How does the median regression estimate compare with OLS?
2. Why might the two estimates differ?
3. What does median regression estimate that OLS does not?

Answers

1. The median regression coefficient is similar to, but not identical to, the OLS estimate.
2. The two estimates differ because the conditional distribution of food expenditure is not perfectly symmetric.
3. Median regression estimates the conditional median rather than the conditional mean.

2.6 Exercise 5: Multiple Quantiles

```

taus <- c(
  0.10,
  0.25,
  0.50,
  0.75,
  0.90
)

fit_multi <- rq(
  foodexp ~ income,
  tau = taus,
  data = engel
)

coef(fit_multi)

```

	tau= 0.10	tau= 0.25	tau= 0.50	tau= 0.75	tau= 0.90
(Intercept)	110.1415742	95.4835396	81.4822474	62.3965855	67.3508721
income	0.4017658	0.4741032	0.5601806	0.6440141	0.6862995

Questions

1. Is the income coefficient constant across quantiles?
2. Which quantile has the largest coefficient?
3. What does this imply about spending behaviour?

Answers

1. No. The income coefficient varies across quantiles.
2. The largest coefficient occurs at the 90th quantile (approximately 0.686).
3. Income has a stronger effect among households with higher food expenditure.

2.6.1 Formal Test of Equality Across Quantiles

```
anova(fit_multi)
```

Quantile Regression Analysis of Deviance Table

Model: foodexp ~ income

Joint Test of Equality of Slopes: tau in { 0.1 0.25 0.5 0.75 0.9 }

	Df	Resid	Df	F value	Pr(>F)
1	4	1171	11.466	3.927e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Questions

4. What is the null hypothesis being tested?
5. What does the p-value suggest?
6. Is there evidence that the income effect differs across quantiles?

Answers

4. The null hypothesis is that the income coefficient is identical across all quantiles.
5. A small p-value indicates evidence against slope homogeneity.
6. Yes. The data suggest that the effect of income differs across the food expenditure distribution.

2.7 Exercise 6: Confidence Intervals

```
income_df <- map_dfr(taus, \(t){

  fit <- rq(
    foodexp ~ income,
    tau = t,
    data = engel
  )

  s <- summary(
    fit,
    se = "nid"
  )

  # Alternatives:
```

```

# s <- summary(fit, se = "boot", R = 50)    # quick experimentation
# s <- summary(fit, se = "boot", R = 1000)  # more stable bootstrap inference

cm <- s$coefficients

tibble(
  tau   = t,
  est   = cm["income", "Value"],
  lower = cm["income", "Value"] -
    1.96 * cm["income", "Std. Error"],
  upper = cm["income", "Value"] +
    1.96 * cm["income", "Std. Error"]
)
})

income_df

```

```

# A tibble: 5 x 4
   tau   est lower upper
<dbl> <dbl> <dbl> <dbl>
1  0.1  0.402 0.323 0.481
2  0.25 0.474 0.417 0.531
3  0.5  0.560 0.505 0.616
4  0.75 0.644 0.598 0.690
5  0.9  0.686 0.630 0.742

```

Questions

1. Are all estimated income effects positive?
2. How do the confidence intervals change across quantiles?
3. Do the intervals suggest that the effect of income is constant?
4. Why are confidence intervals useful when interpreting quantile regression results?

Answers

1. Yes. All estimated income effects are positive.
2. The confidence intervals overlap but are centred at different values.
3. The estimates suggest clear heterogeneity across quantiles. The income coefficient increases steadily from approximately 0.40 at the 10th percentile to approximately 0.69 at the 90th percentile.
4. Confidence intervals quantify uncertainty and help assess whether apparent differences are statistically meaningful.

2.8 Exercise 7: Coefficient Plot

```

ols_income <- coef(ols_fit)["income"]

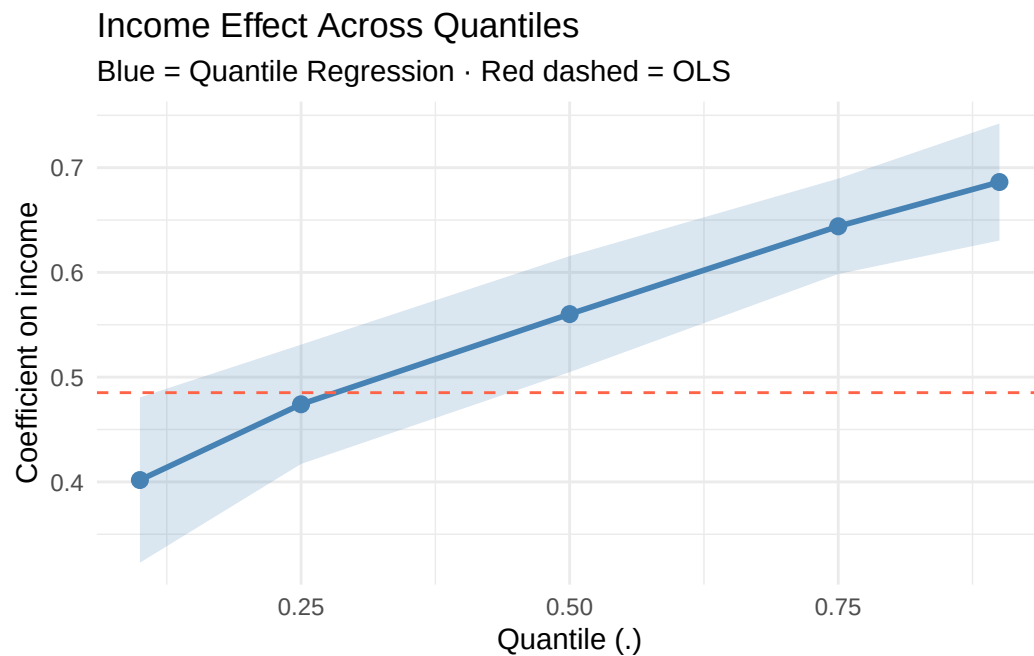
ggplot(
  income_df,
  aes(
    x = tau,
    y = est
  )
) +
  geom_ribbon(
    aes(

```

```

    ymin = lower,
    ymax = upper
  ),
  alpha = 0.20,
  fill = "steelblue"
) +
geom_line(
  colour = "steelblue",
  linewidth = 1
) +
geom_point(
  colour = "steelblue",
  size = 2.5
) +
geom_hline(
  yintercept = ols_income,
  linetype = "dashed",
  colour = "tomato"
) +
labs(
  title = "Income Effect Across Quantiles",
  subtitle = "Blue = Quantile Regression · Red dashed = OLS",
  x = "Quantile (.)",
  y = "Coefficient on income"
)

```



Questions

1. What pattern do you observe?
2. Does the effect of income appear constant?
3. Does OLS adequately summarise the relationship?

Answers

1. The coefficient tends to increase as the quantile increases.
2. No. The effect of income is not completely constant across the distribution.
3. OLS provides a useful average summary, but it masks differences between lower- and higher-spending households.

2.9 Exercise 8: Interpretation

Complete the following paragraph.

At the 10th quantile, an additional unit of income is associated with approximately _____ additional units of food expenditure. At the 90th quantile, the corresponding estimate is _____. This suggests that _____ . The OLS estimate of _____ provides _____ .

Model Answer

At the 10th quantile, an additional unit of income is associated with a smaller increase in food expenditure than at the 90th quantile. This suggests that higher-spending households devote a larger share of additional income to food expenditure than lower-spending households. The OLS estimate provides an average effect across all households but masks this distributional heterogeneity.

Reflection

1. What did quantile regression reveal that OLS could not?
2. Which result was most surprising?
3. In what other applications might quantile regression be useful?

Possible Answers

1. Quantile regression revealed that the effect of income is not identical across the food expenditure distribution.
2. Answers will vary.
3. Examples include wages and education, housing prices, health outcomes, consumption behaviour, and programme evaluation.